

BUYU DENG

School of the Gifted Young, University of Science and Technology of China
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EDUCATION

University of Science and Technology of China

B.S. Candidate in Geophysics, School of the Gifted Young
GPA: 3.62/4.30; rank: 3/20

Hefei, China

Expected June 2027

Coursework: Programming in Geophysics (94), Intro. to Seismic Exploration (98), Geophysical Data Processing and Analysis (93), Linear Algebra (90), Probability & Statistics (92), Theoretical Mechanics (95), Python & Deep Learning (A).

RESEARCH INTERESTS

AI for exploration geophysics; seismic interpretation, geomodeling, rock physics, and forward modeling.

RESEARCH EXPERIENCE

Stanford Center for Earth Resources Forecasting (SCERF), Stanford University

Undergraduate Research Intern; advisor: Prof. Tapan Mukerji

Stanford, CA

July – October 2026

CO₂ Storage Modeling and Rock Physics Agent Development

- Compared five vertical grid resolutions (0.04–0.92 ft) in ECLIPSE 300 CO₂STORE simulations using fixed-time plume geometry, fixed-location saturation histories, space-time plume metrics, and computational cost; selected 0.12 ft for subsequent modeling.
- Built a facies-based rock physics workflow using mineral averaging, Hertz–Mindlin dry frames, Wood fluid mixing, and Gassmann substitution to map geology and CO₂ saturation to elastic properties.
- Generated baseline and time-lapse seismic responses for 20 years of CO₂ injection and 80 years after shut-in; interpreted horizon-relative amplitude slices in Petrel.
- Designed an agent workflow around the Stanford Rock Physics and Borehole Geophysics (SRB) model library exposed through Model Context Protocol (MCP) tools for model discovery, input validation, and traceable calculations.
- Conducted 45 independent agent runs across Smeaheia, glass-bead, and White patchy-saturation benchmarks, comparing access to textbook guidance and SRB tools.
- Compared three resource settings: without references, agents made unsupported model assumptions; textbook guidance recovered the prescribed physics and reference results; textbook plus SRB tools enabled direct model calls but required careful interpretation of outputs.

Computational Interpretation Group (CIG), University of Science and Technology of China

Undergraduate Researcher; advisor: Prof. Xinming Wu

Hefei, China

August 2024 – Present

Geo-VAE for 3D Geophysical Data Compression and Latent Processing

- Adapted Wan-VAE with cached causal pseudo-temporal resampling along the crossline axis, achieving isotropic $8 \times 8 \times 8$ downsampling and 31.0:1 scalar compression for 256^3 cubes.
- Demonstrated strong reconstruction fidelity across five 3D geophysical modalities while achieving a favorable balance of GPU memory use and inference time among tested state-of-the-art video VAEs on 128^3 – 512^3 cubes.
- Built a unified compact latent space for RGT-conditioned seismic generation and complete-volume 768^3 denoising, enabling condition-responsive generation and scalable field-volume processing.

Code: github.com/BuyuDeng/Geo-VAE

RESEARCH EXPERIENCE (CONTINUED)

3D All-in-One Seismic Denoising

- Built a synthetic 3D seismic dataset with random noise, acquisition footprints, and migration artifacts; trained an all-in-one denoising network based on Degradation Representation Learning.

Code: github.com/BuyuDeng/NDR_Seismic

ADDITIONAL RESEARCH PROJECTS

PHY-to-BGC Ocean Biogeochemical Emulation

- Built a 2.5D full-image U-Net emulator mapping global physical ocean fields to coupled biogeochemical variables, including evaluation under anomalous ocean-climate conditions. Manuscript in preparation.

GNSS–NISAR Ionospheric Super-Resolution Reconstruction

- Developed an automated pipeline aligning GNSS TEC-derived ionospheric phase with NISAR dual-frequency InSAR for high-resolution reconstruction research.

PUBLICATIONS AND PRESENTATIONS

B. Deng, Y. Dou, Z. Guo, and X. Wu. "Geo-VAE: A Unified Variational Autoencoder for 3D Geophysical Data Compression and Latent Processing." Submitted to *Geophysics*; major revision requested, 2026.

B. Deng. "Geo-VAE: A Unified Variational Autoencoder for 3D Geophysical Data Compression and Latent Processing." Oral presentation, International Meeting for Applied Geoscience & Energy (IMAGE 2026), Houston, TX, August 2026.

B. Deng. "CO₂ Storage Modeling and Rock Physics Agent Development." SCERF Seminar, Stanford University, September 2026.

Y. Dou, X. Wu, Z. Guo, H. Gao, and **B. Deng**. "Implicit Structural Modeling via Generative Diffusion Frameworks." *arXiv:2606.07165*, 2026. arxiv.org/abs/2606.07165.

RESEARCH SKILLS

Computing: Python, deep learning for 3D geophysical data, numerical modeling, scientific visualization.

Geophysics: Seismic interpretation and forward modeling, rock physics, CO₂ flow simulation, ECLIPSE 300, Petrel.

HONORS AND AWARDS

Zhao Jiuzhang Elite Program Scholarship, School of Earth and Space Sciences	2025
Rose Fund Endeavor Scholarship, School of the Gifted Young	2024
Zhao Jiuzhang Elite Program Scholarship, School of Earth and Space Sciences	2024
Freshman Scholarship, University of Science and Technology of China	2023

FIELD EXPERIENCE AND LEADERSHIP

IPGP Geophysical Field Camp, Institut de Physique du Globe de Paris August 2025
Conducted geophysical surveys and joined academic seminars with IPGP faculty.

Head (Minister), Department of Cultural & Sports, College Student Union June 2025 – January 2026